

Assessing Unconfoundedness in Two Cases*

Contents

1	Energy Conservation Field Experiments	2
1.1	Background	2
1.2	The Field Experiment Design	2
1.3	The Recruitment and Assignment of Consumers	3
1.4	Challenges in Data Analysis	3
2	A Quasi-experiment of Online Reviews	5
2.1	Background	5
2.2	Study Design	6
2.3	Evaluating the Validity of the Design	6

*This case is prepared solely for Causal Inference at GSERM.

Please do NOT circulate.

1 Energy Conservation Field Experiments

1.1 Background

Following consultation with industry, the Commission for Energy Regulation (CER) in Ireland established the Phase I Smart Metering Project in late 2007, powered by the installation of smart meters in all of Ireland. From smart meters, the CER has access to granular data on the electricity use of households. CER records electricity usage at half-hour intervals for each home with a smart meter installed. Around March 2008, CER began to prepare a Customer Behavior Trial to ascertain the potential for smart metering technology, when combined with time-of-use tariffs and different Demand-Side Management (DSM) stimuli to effect measurable change in consumer behavior.

1.2 The Field Experiment Design

The experiment focused on the time-of-use tariffs and DSM stimuli and their possible interactions. These were specifically developed for use in the customer behavior test.

- Four specific time-of-use tariffs (A, B, C, D) offering different unit prices for night, day, and peak times.
- Specific DSM initiatives, including:
 - Bi-monthly electricity bill with detailed energy statement
 - Monthly electricity bill with detailed energy statement
 - Electricity in-home display
 - Overall Load Reduction (OLR) incentive

Two key design features:

- Tariffs were neutral compared to the standard regulated tariff, so participants who did not alter usage were not penalized.
- Relative differences among low vs normal vs. peak rates varied between tariffs (e.g., tariff D: 38.00 vs. 9.00; tariff A: 20.00 vs. 12.00).

The DSM stimuli varied the frequency of energy statements (monthly vs. bi-monthly), provided in-home energy displays, and included monetary incentives.

1.3 The Recruitment and Assignment of Consumers

Recruitment required consumer consent: out of 5,028 contacted, 3,858 opted in for treatment, and 1,170 were assigned to the control group. Treatment assignments used stratified block randomization based on analysis of demographic and electrical usage factors.

1.4 Challenges in Data Analysis

The final data set includes half-hour electricity usage during the pre- (July-Dec. 2009) and post-treatment (Jan.-Dec. 2010) periods. A difference-in-differences (DID) design was adopted to account for confounders that are stable over time (fixed individual effects) and those common between consumers but varying over time (fixed time effects).

- DID design relies on the assumption of parallel trends (unconfoundedness).
- Although untestable, implications of the assumption can be explored with the data and design structure.

Energy awareness



Typical cost of running various appliances over a full year*

Main household appliances (excl. Electric Oven)	NIGHT RATE	DAY RATE	PEAK RATE
Washing machine	€55	€64	€91
Tumble dryer	€183	€213	€305
Dishwasher	€73	€85	€122
Immersion - 6 months only	€203	€236	€338

* Average usage 1 cycle per day, 5 days a week for a full year. Immersion: 1 tank per day 6 months only.

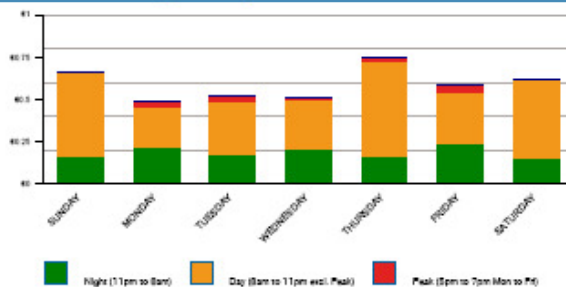
Hints and Tips

- Money Down the Drain - During the peak period (5pm to 7pm) an instantaneous electric shower running for 15 minutes costs you €114.30 per year. At day rates it would cost you €80.01 per year.
- Beat the Peak. If using a dishwasher washing your dishes at peak rate will cost you €121.92 per year; on the day rate it would cost you €85.34 per year.
- Remember! To make the greatest savings always try to be energy efficient when you use your appliances even when you shift to an Off-Peak time.

Has your electricity usage changed?

- 2.8% of your electricity for last month was used in the peak period. By using some of your appliances at day rate rather than at peak rate you could save money.
- Last month 572 customers on your tariff have reduced the amount of electricity they use. You are one of them. Congratulations!

Your average day of the week costs



Learn More

2010 Electricity Reduction Target

Your Target

Use less than **330** Units By 31 Dec

Used so far **330** Units

Reduce your usage Goals

Goal THIS bill - Use less than **330** Units

Goal NEXT bill - Use less than **330** Units

You are outside your goal for this period, can you reduce your usage more? See the energy tips above.

Your goals help to meet your annual target. Reach your target in December and earn €20 AS WELL as any savings you make on your bill.

Further information

Values given above may be slightly different to Page 1 due to rounding impacts. The correct final values are those displayed on Page 1 of the Bill.

Figure 1: Energy Usage Statement Example



Figure 2: In-home Display of Energy Usage

Composition of Eigen vectors	Eigen Vectors				
Factors	V1	V2	V3	V4	V5
Peak band	-0.178	0.577	-0.065	0.085	-0.087
Night band	0.103	-0.526	0.101	0.142	0.153
Overall Weekly Usage Variance	-0.039	0.1	0.133	0.164	0.485
Overall Peak Usage Variance	-0.152	0.53	0.01	0.193	0.207
Number in house	-0.395	-0.003	-0.11	-0.288	-0.276
# of bedrooms	-0.312	-0.119	-0.078	-0.247	-0.197
Internet access	-0.414	-0.119	0.006	0.095	-0.078
Income band	-0.463	-0.12	0.043	0.039	0.094
Education classification	-0.361	-0.178	0.071	0.196	0.288
Employment status	-0.401	-0.087	0.036	0.1	0.125
Wet	0.013	0.006	0.028	0.582	-0.304
Electronics	-0.004	-0.092	0.044	0.568	-0.398
Energy reduction engagement band	0.022	-0.047	-0.646	0.08	0.103

Figure 3: Factor Loadings of Different Variables

2 A Quasi-experiment of Online Reviews

2.1 Background

Consumer review platforms like Yelp.com provide valuable information for experience products. However, endogeneity issues pose a challenge in identifying the causal effect

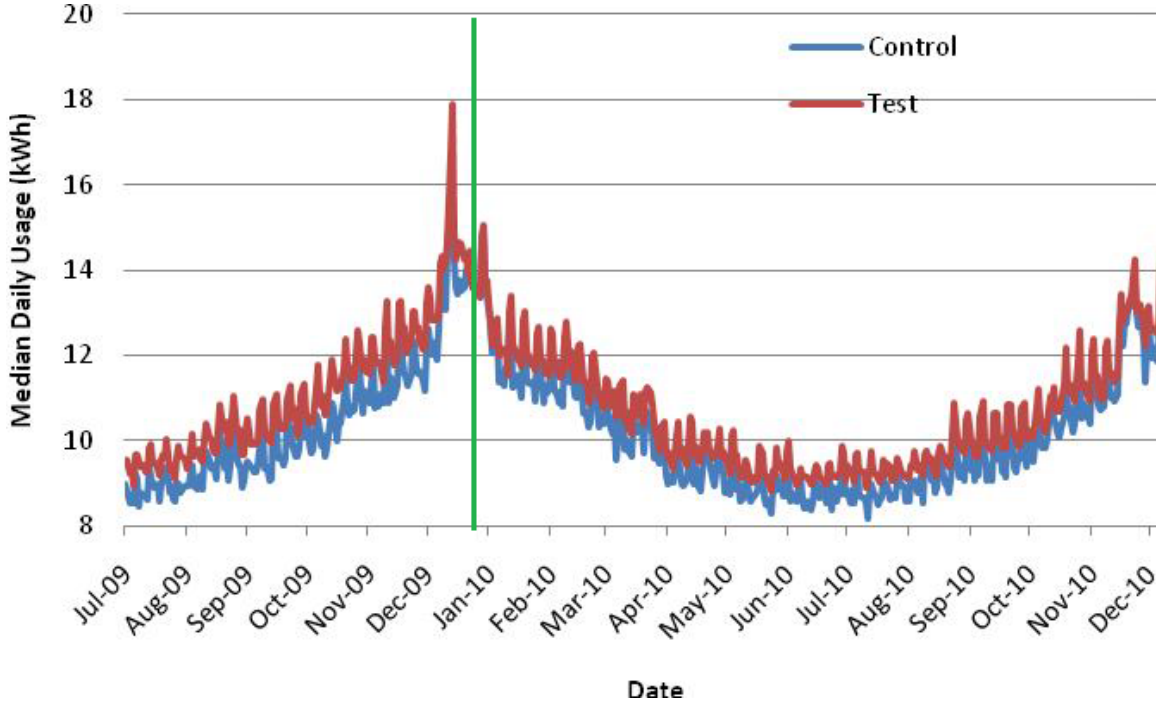


Figure 4: Median Daily Usage (kWh) of the Test and Control Group

of reviews on market outcomes. For example, unobserved marketing efforts might be correlated with both review ratings and consumer behavior.

2.2 Study Design

Yelp data scientists studied how review ratings influence restaurant revenues (2003–2009). They exploited Yelp’s rounding of average ratings (to the nearest half-star) to identify discontinuities that could be treated as exogenous.

They used a regression discontinuity (RD) design to examine revenue jumps caused by rounding-induced rating changes.

2.3 Evaluating the Validity of the Design

A key threat to the identification of RDs is manipulation or gaming near the threshold (McCrary, 2008). In the case of Yelp, this would imply that some restaurants may submit false reviews to increase ratings.

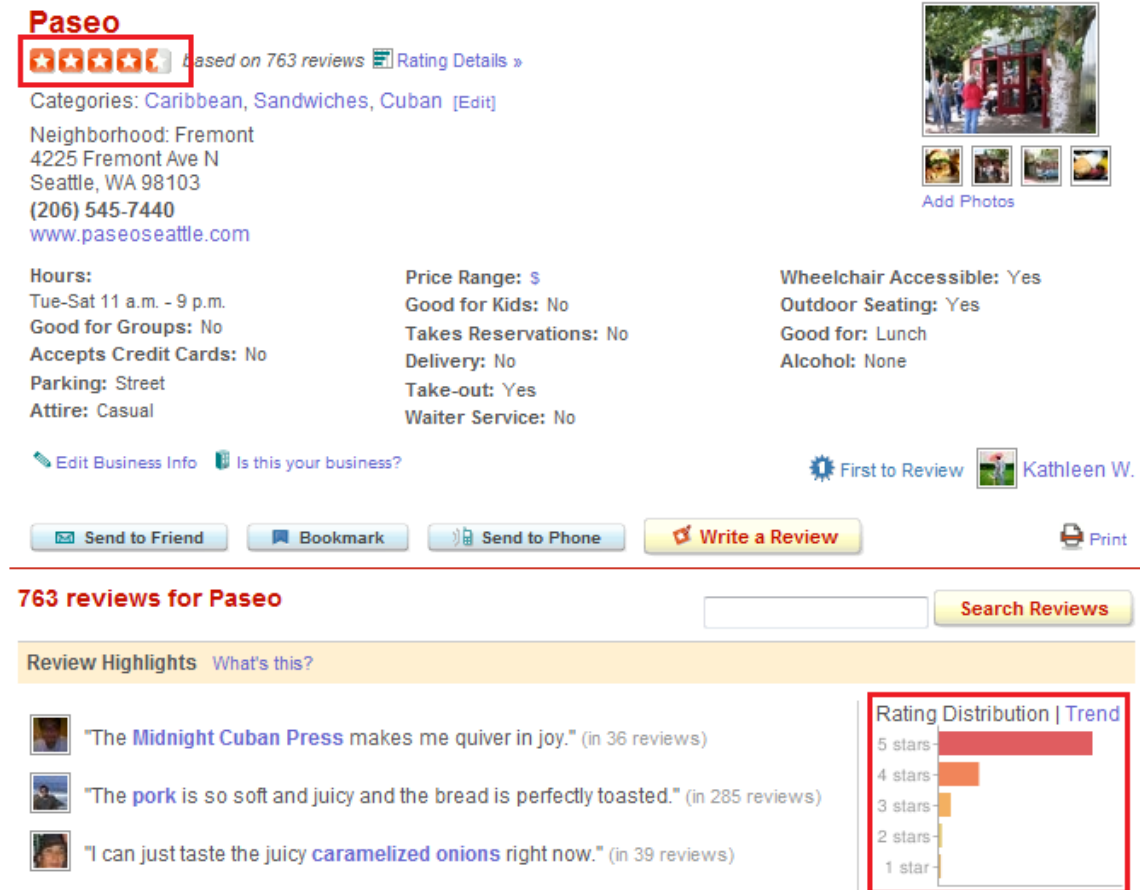


Figure 5: Yelp Page Showing a Restaurant Rated 4.5

Questions to explore include

- How likely is manipulation in this setting?
- What tests (e.g., McCrary density test) can be applied to detect it using Yelp data?

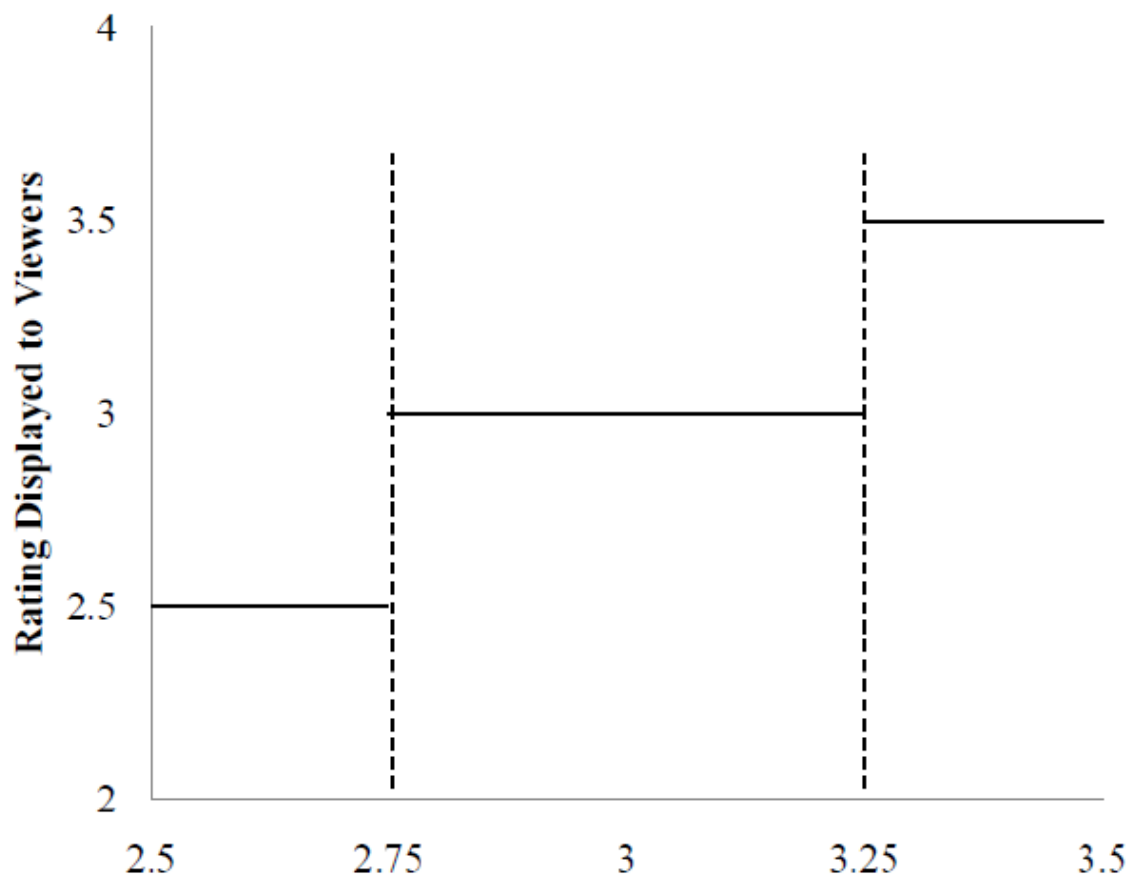


Figure 6: Yelp Displays Each Restaurant's Rounded Average Rating

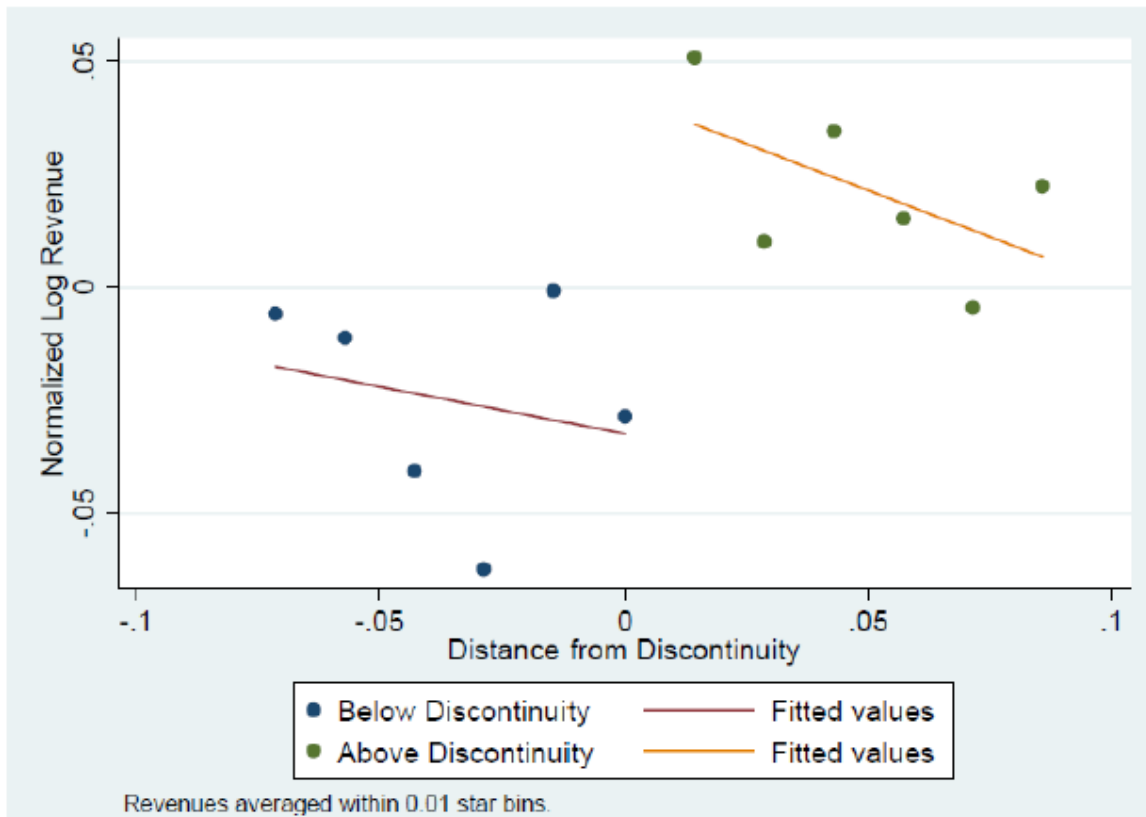


Figure 7: Average Revenue Around Discontinuous Changes in Rating